

Main Content

- Derivation of the Discrete Time Fourier Series (DTFS)
- Properties of DTFS

1. Derivation of the Discrete Time Fourier Series (DTFS)

(1) Continuous Fourier Series Recap

A continuous periodic signal $\tilde{x}(t)$ with period T can be expressed as:

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

where:

$\omega_0 = \frac{2\pi}{T}$ (fundamental frequency),

a_k are the Fourier coefficients:

$$a_k = \frac{1}{T} \int_0^T \tilde{x}(t) e^{-jk\omega_0 t} dt$$

(2) Moving to Discrete-Time Signals

For a discrete-time periodic signal $\tilde{x}[n]$ with period N , we sample $\tilde{x}(t)$ at intervals $t = nT_s$, where T_s is the sampling period

The discrete-time signal is periodic:

$$\tilde{x}[n] = \tilde{x}[n + mN] \text{ for any integer } m$$

The fundamental digital frequency is $\Omega_0 = \frac{2\pi}{N}$

(3) Discrete-Time Fourier Series Representation

We seek to represent $\tilde{x}[n]$ as a sum of complex exponentials:

$$\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{X}[k] e^{jk\Omega_0 n}$$

where:

$\tilde{X}[k]$ are the DFS coefficients(discrete frequency components),

The summation is over N terms because the signal is periodic in N

(4) Orthogonality of Discrete Exponentials

The key to finding $\tilde{X}[k]$ is the orthogonality of discrete complex exponentials:

$$\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-r)n} = \begin{cases} N & \text{if } k = r \\ 0 & \text{otherwise.} \end{cases}$$

This means:

If $k \neq r$, the sum is zero.

Exponentials are distinct and orthogonal; Analogous to how sine/cosine waves of different frequencies integrate to zero over a period

If $k = r$, the sum is N .

(5) Solving for the DFS Coefficients $X[k]$

Multiply both sides of the DFS representation by $e^{-jr\Omega_0 n}$ and sum over one period N :

$$\sum_{n=0}^{N-1} \tilde{x}[n] e^{-jr\Omega_0 n} = \sum_{n=0}^{N-1} \left(\sum_{k=0}^{N-1} \tilde{X}[k] e^{jk\Omega_0 n} \right) e^{-jr\Omega_0 n}$$

Using orthogonality, the right-hand side simplifies to:

$$\sum_{k=0}^{N-1} \tilde{X}[k] \sum_{n=0}^{N-1} e^{j(k-r)\Omega_0 n} = N \tilde{X}[r] \quad k = r$$

Thus, the DFS analysis equation is:

$$\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$$

(6) Summary of DFS Equations

Synthesis (Inverse DFS):

$$\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}$$

Analysis (Forward DFS):

$$\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$$

2. Properties of the DFS

- Periodicity
- Linearity
- Shifting
- Symmetry
- Parseval' s Theorem
- Periodic/Circular Convolution

(1) Periodicity

Time Domain: The signal $\tilde{x}[n]$ is periodic with period N :

$$\tilde{x}[n + N] = \tilde{x}[n] \quad \forall n$$

Frequency Domain: The DFS coefficients $X[k]$ are also periodic with period N :

$$\tilde{X}[k + N] = \tilde{X}[k] \quad \forall k$$

Implication: Both the signal and its spectrum repeat every N samples

(2) Linearity

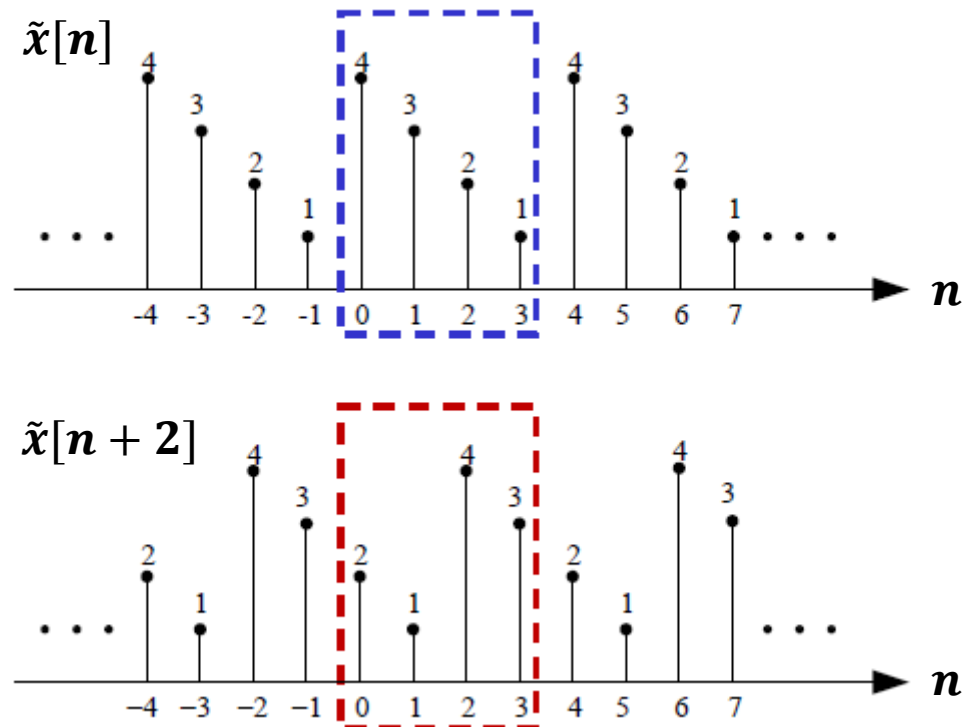
For two periodic signals $\tilde{x}_1[n]$ and $\tilde{x}_2[n]$ with DFS coefficients $\tilde{X}_1[k]$ and $\tilde{X}_2[k]$, and constants a and b :

$$a\tilde{x}_1[n] + b\tilde{x}_2[n] \xrightarrow{DFS} a\tilde{X}_1[k] + b\tilde{X}_2[k]$$

Implication: DFS preserves linear combinations, making it useful for superposition-based analysis

6. Discrete Time Fourier Series

(3) Shifting



A shift in the time domain still results in a periodic sequence with the same period. Thus, it is sufficient to analyze the shifted sequence over one period

(3) Shifting

Time Shifting

Shifting the signal by n_0 samples modulates the DFS coefficients:

$$\tilde{x}[n - n_0] \leftrightarrow \tilde{X}[k]e^{-j\frac{2\pi}{N}kn_0}$$

Implication: A **delay in time** introduces a linear **phase shift** in the **frequency domain**

(3) Shifting

Frequency Shifting

Modulating the signal by a complex exponential shifts its spectrum:

$$\tilde{X}[k - k_0] \leftrightarrow \tilde{x}[n] e^{j\frac{2\pi}{N}k_0n}$$

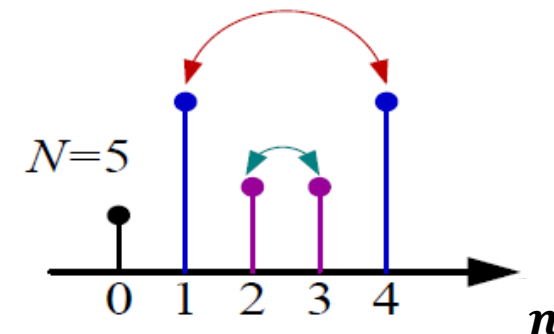
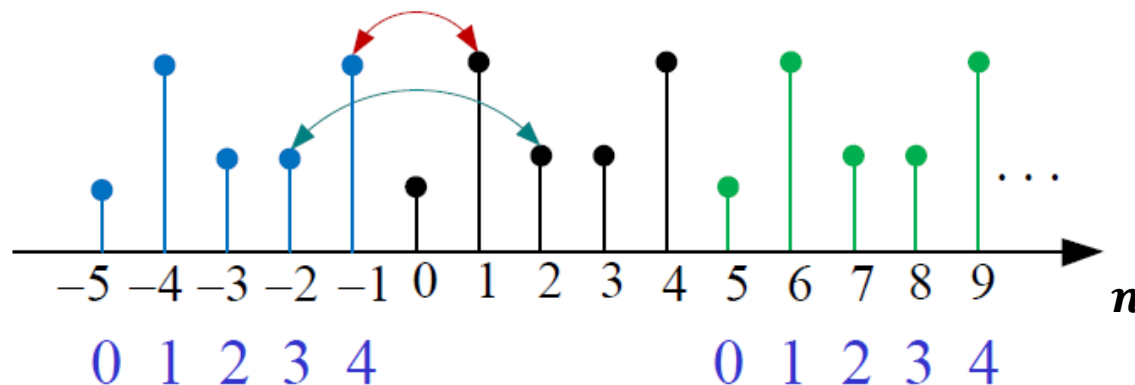
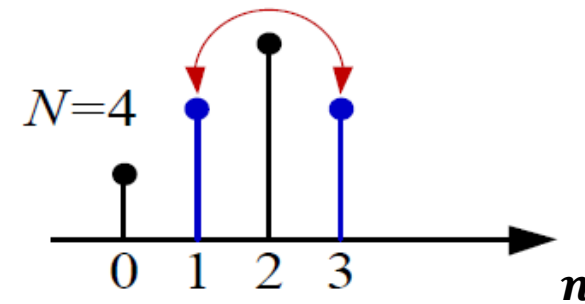
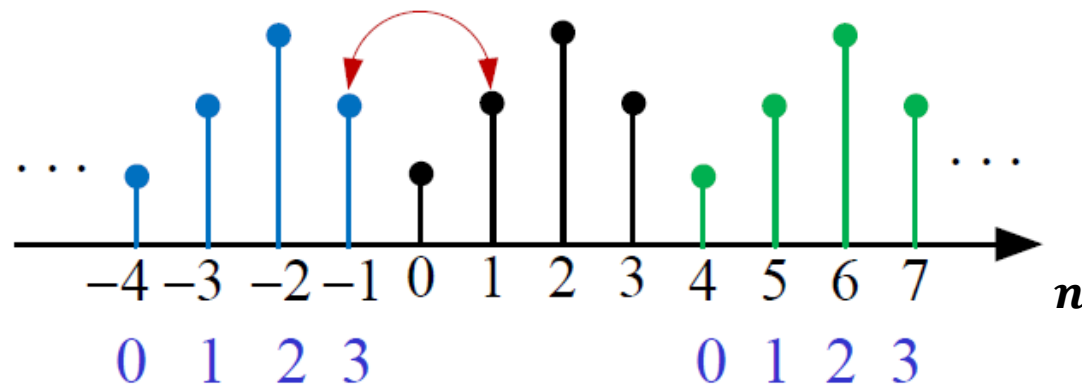
A modulation in the time-domain corresponds to a frequency-domain displacement

Implication: Used in modulation techniques (e.g., AM/FM)

6. Discrete Time Fourier Series

(4) Symmetry

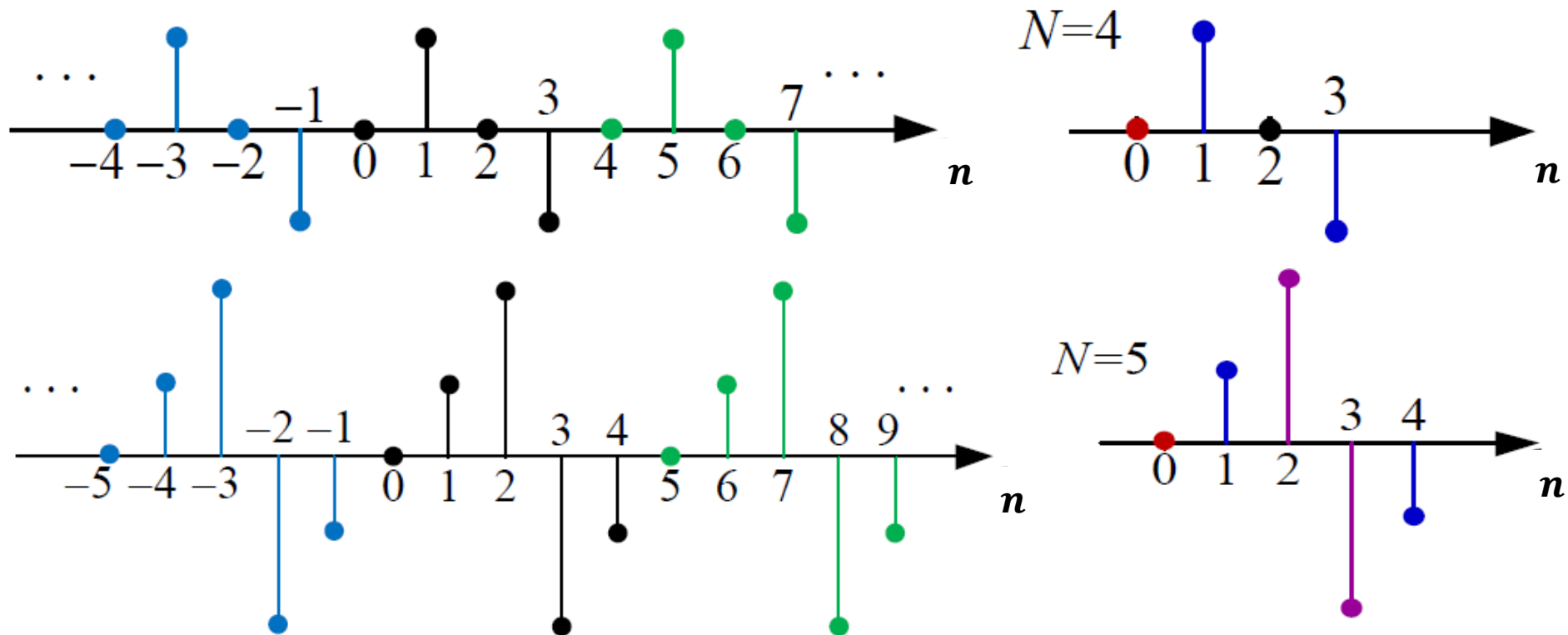
Even Symmetry in
Periodic Sequences : $\tilde{x}[n] = \tilde{x}[-n] = \tilde{x}[N - n]$



6. Discrete Time Fourier Series

(4) Symmetry

Odd Symmetry in Periodic Sequences : $\tilde{x}[n] = -\tilde{x}[-n] = -\tilde{x}[N - n]$



(4) Symmetry

If

$$\tilde{x}[n] \xleftrightarrow{\text{DFS}} \tilde{X}[k],$$

Then

1. Conjugation in time:

$$\tilde{x}^*[n] \xleftrightarrow{\text{DFS}} \tilde{X}^*[-k]$$

2. Conjugation & reversal in time:

$$\tilde{x}^*[-n] \xleftrightarrow{\text{DFS}} \tilde{X}^*[k]$$

(5) Parseval's Theorem

Relates the total **energy** in the time and frequency domains for one N -point period:

$$\sum_{n=0}^{N-1} |\tilde{x}[n]|^2 = N \sum_{k=0}^{N-1} |\tilde{X}[k]|^2$$

Implication: Energy is conserved between domains (up to scaling factor N), and $|\tilde{X}[k]|^2$ represents the energy at frequency bin k , $1/N |\tilde{X}[k]|^2$ represents the **power** at frequency bin k

(6) Circular Convolution

Circular Convolution: **Multiplication** in one domain corresponds to **circular convolution** in the other:

$$\tilde{x}[n] \circledast \tilde{h}[n] \xrightarrow{DFS} N \cdot \tilde{X}[k] \cdot \tilde{H}[k]$$

$$\tilde{x}[n] \cdot \tilde{h}[n] \xrightarrow{DFS} \tilde{X}[k] \circledast \tilde{H}[k]$$

Here, $\circledast$ denotes circular convolution

Implication: Efficient for filtering periodic signals (e.g., in digital communications)

Main Content

- Discrete Time Fourier Transform (DTFT)
- Discrete Fourier Transform (DFT)
- Properties of DFT

1. Discrete Time Fourier Transform (DTFT)

Start with the Fourier Transform for Continuous Signals: The continuous-time Fourier Transform (CTFT) is:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

For discrete-time signals, replace t with nT (sampling interval T) and the integral with a sum:

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

1. Discrete Time Fourier Transform (DTFT)

The DTFT of a sequence $x[n]$ is given by:

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

where:

- Ω is the normalized angular frequency ($\Omega = 2\pi f$)
- $X(e^{j\Omega})$ is periodic with period 2π

Inverse DTFT (IDTFT):

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

1. Discrete Time Fourier Transform (DTFT)

- **Periodicity of DTFT:** Since $e^{-j\Omega n} = e^{-j(\Omega+2\pi)n}$, the DTFT is periodic with 2π
- **Convergence Conditions:** The DTFT exists if $x[n]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

2. Discrete Fourier Transform (DFT)

Deriving the DFT from the DTFT

Step 1: Truncate the Signal

The DTFT assumes an infinite-length sequence. To make it practical:

- Consider a finite-length sequence $x[n]$ defined for $n = 0, 1, \dots, N - 1$
- For all other n , $x[n] = 0$

Step 2: Sample the DTFT in Frequency

The DTFT $X(e^{j\Omega})$ is continuous in Ω . To discretize it:

- Evaluate $X(e^{j\Omega})$ at equally spaced frequencies:

$$\Omega_k = \frac{2\pi k}{N}, k = 0, 1, \dots, N - 1$$

- Substituting into the DTFT:

$$X[k] = X(e^{j\Omega_k}) = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

**Frequency
domain sampling**

2. Discrete Fourier Transform (DFT)

Transform	Domain	Frequency Representation	Key Characteristics
DTFT	Infinite-length sequences	Continuous ($\Omega \in [-\pi, \pi]$)	$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$
DFT	Finite-length sequences (N samples)	Discrete ($k = 0, 1, \dots, N - 1$)	$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$

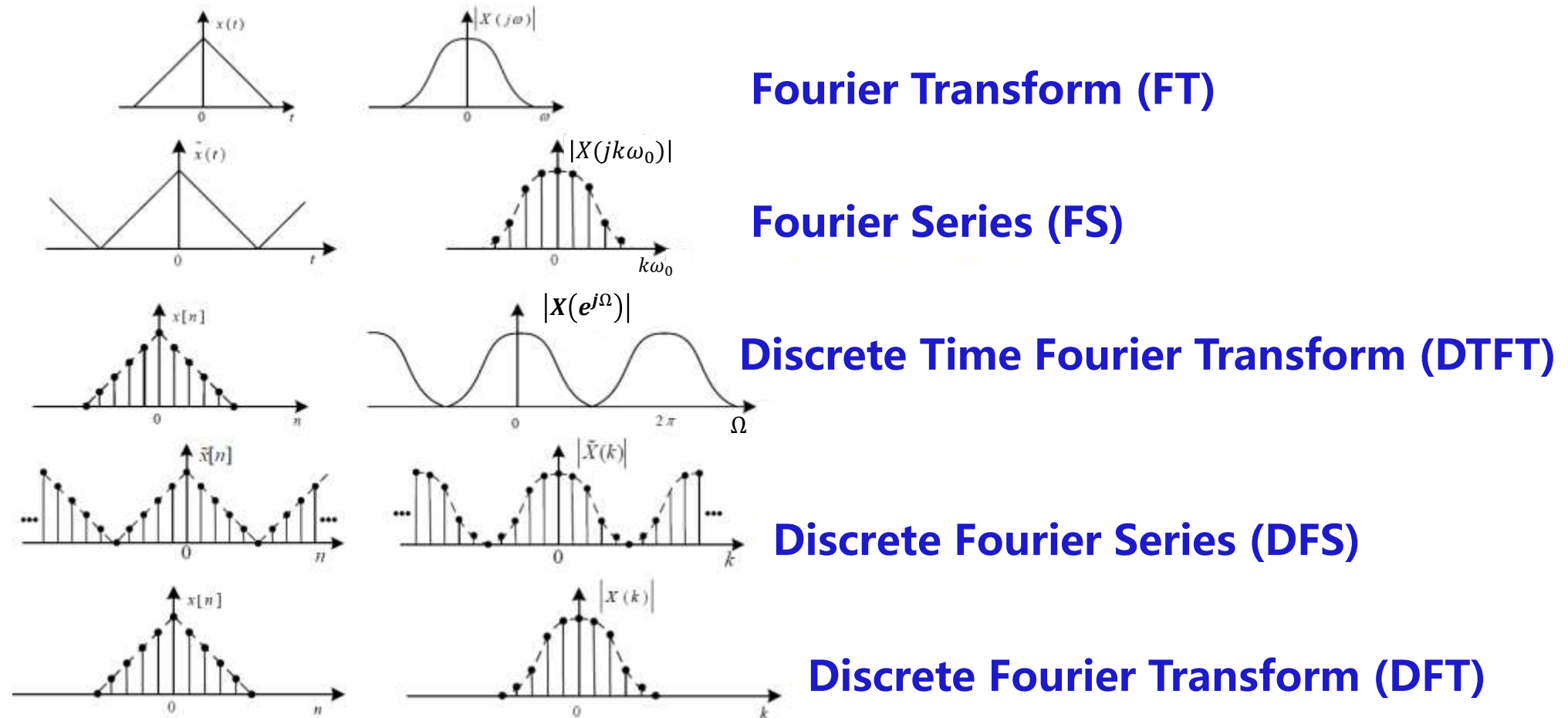
7. DTFT and DFT

Fourier Transform Family: Complete Comparison

Transformation	Time domain characteristics	Frequency domain characteristics
Fourier Transform (FT)	Continuous, Aperiodic $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$	Continuous, Aperiodic $x(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$
Fourier series (FS)	Continuous, Periodic $\tilde{x}(t) = \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t}$	Discrete, Aperiodic $X(jk\omega_0) = \frac{1}{T} \int_0^T \tilde{x}(t) e^{-jk\omega_0 t} dt$
Discrete-time Fourier transform (DTFT)	Discrete, Aperiodic $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$	Continuous, Periodic $X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$
Discrete Fourier series (DFS)	Discrete, Periodic $\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{X}(k) e^{j\frac{2\pi}{N}kn},$ $-\infty < n < +\infty$	Discrete, Periodic $\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}kn},$ $-\infty < k < +\infty$
Discrete Fourier transform (DFT)	Discrete, Implicit Periodic $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N}kn},$ $0 \leq n \leq N-1$	Discrete, Implicit Periodic $X[k] = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn},$ $0 \leq k \leq N-1$

7. DTFT and DFT

Fourier Transform Family: Complete Comparison



DFT is DFS evaluated over a single period (**principal value interval**) of a finite-length sequence

2. Discrete Fourier Transform (DFT)

The definition formula for the discrete Fourier transform of a finite time domain (length N) discrete time series $x(n)$ is as follows:

$$\begin{aligned} X(k) &= \text{DFT}[x(n)] \\ &= \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}nk} \quad 0 \leq k \leq N-1 \end{aligned} \quad W = e^{-j\frac{2\pi}{N}} \quad X(k) = \sum_{n=0}^{N-1} x(n) W^{nk}$$

The corresponding inverse transformation of discrete Fourier transform is

$$\begin{aligned} x(n) &= \text{IDFT}[X(k)] \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N}nk} \quad 0 \leq n \leq N-1 \end{aligned} \quad x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W^{-nk}$$

k is the index on the frequency domain, while n is the index on the time domain, with N is the length of sequence $0 \sim (N-1)$

W is the **twiddle** factor (**phase rotation** factor)

2. Discrete Fourier Transform (DFT)

$$\begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix} = \begin{bmatrix} W^0 & W^0 & W^0 & \dots & W^0 \\ W^0 & W^{1 \times 1} & W^{2 \times 1} & \dots & W^{(N-1) \times 1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W^0 & W^{1 \times (N-1)} & W^{2 \times (N-1)} & \dots & W^{(N-1) \times (N-1)} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix} = \frac{1}{N} \begin{bmatrix} W^0 & W^0 & W^0 & \dots & W^0 \\ W^0 & W^{-1 \times 1} & W^{-2 \times 1} & \dots & W^{-(N-1) \times 1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W^0 & W^{-1 \times (N-1)} & W^{-2 \times (N-1)} & \dots & W^{-(N-1) \times (N-1)} \end{bmatrix} \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$$

Example 1

1. Calculate the DFT of a finite length sequence $\{h(n)\} = \{1, 2, -1, 3\}$

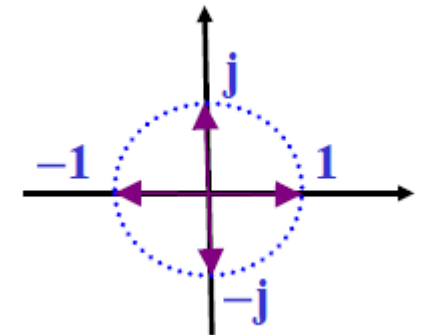
Solve: $H(k) = \sum_{n=0}^3 h(n) W_4^{nk}, \quad N = 4$

$$\begin{bmatrix} H(0) \\ H(1) \\ H(2) \\ H(3) \end{bmatrix} = \begin{bmatrix} W_4^{0 \cdot 0} & W_4^{0 \cdot 1} & W_4^{0 \cdot 2} & W_4^{0 \cdot 3} \\ W_4^{1 \cdot 0} & W_4^{1 \cdot 1} & W_4^{1 \cdot 2} & W_4^{1 \cdot 3} \\ W_4^{2 \cdot 0} & W_4^{2 \cdot 1} & W_4^{2 \cdot 2} & W_4^{2 \cdot 3} \\ W_4^{3 \cdot 0} & W_4^{3 \cdot 1} & W_4^{3 \cdot 2} & W_4^{3 \cdot 3} \end{bmatrix} \begin{bmatrix} h(0) \\ h(1) \\ h(2) \\ h(3) \end{bmatrix}$$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = -j, \quad W_4^2 = -1, \quad W_4^3 = W_4^1 = W_4^2 = j, \quad W_4^4 = W_4^0 = 1$$

$$\begin{bmatrix} H(0) \\ H(1) \\ H(2) \\ H(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 + j \\ -5 \\ 2 - j \end{bmatrix}$$

$$\{H(k)\} = \{5, 2 + j, -5, 2 - j\}$$



3. Properties of DFT

- **Linearity**
- **Circular (Periodic) Shift**
- **Circular Convolution**

(1) Linearity

If the linear combination of finite length sequences $x_1(n)$ and $x_2(n)$ is

$$x_3(n) = ax_1(n) + bx_2(n)$$

a, b : constants

$X_1(k)$, $X_2(k)$ and $X_3(k)$ are DFTs of $x_1(n)$, $x_2(n)$ and $x_3(n)$ respectively

$$\text{Then, } X_3(k) = aX_1(k) + bX_2(k)$$

If the lengths of $x_1(n)$ and $x_2(n)$ are N_1 and N_2 respectively, with $N_2 > N_1$, the shorter sequence $x_1(n)$ must be **zero-padded** by appending L zeros to match the length of $x_2(n)$. This ensures $N_2 = N_1 + L$, the resulting sequence $x_3(n)$ will then have a length of N_2

(2) Circular (Periodic) Shift

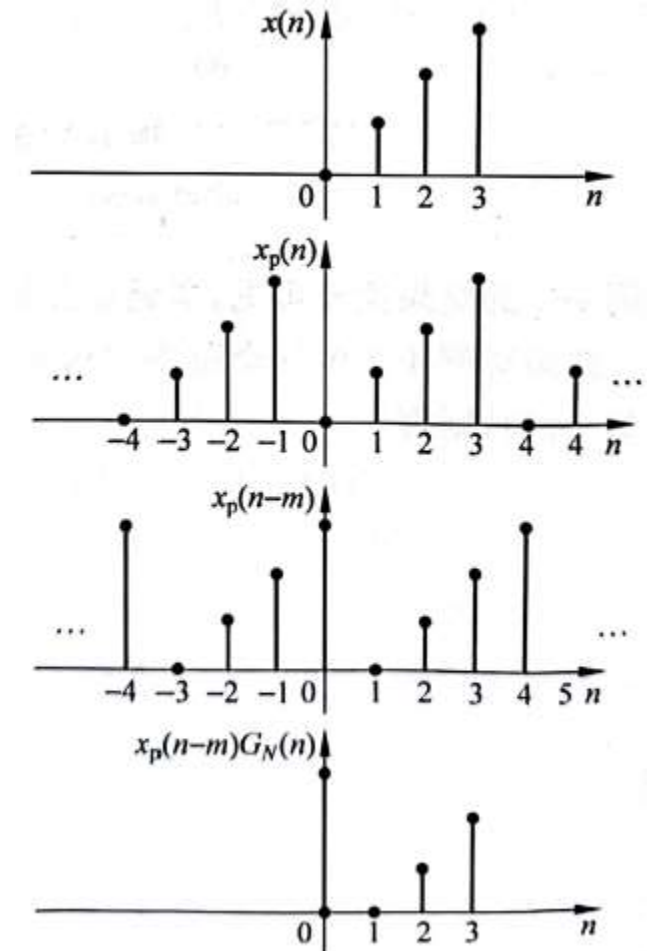
Periodic Extension:

The sequence $x(n)$ with a length of N is periodically extended to generate the periodic sequence $x_p(n)$

Circular Shift by m Samples

$$x_p(n - m)$$

Principal Value Interval Extraction



(2) Circular (Periodic) Shift

Principal Value Interval Extraction

$$x_p(n) = \sum_{r=-\infty}^{\infty} x(n - rN), \quad r \text{ is an integer}$$

$$\text{and, } 0 \leq n - rN \leq N - 1$$

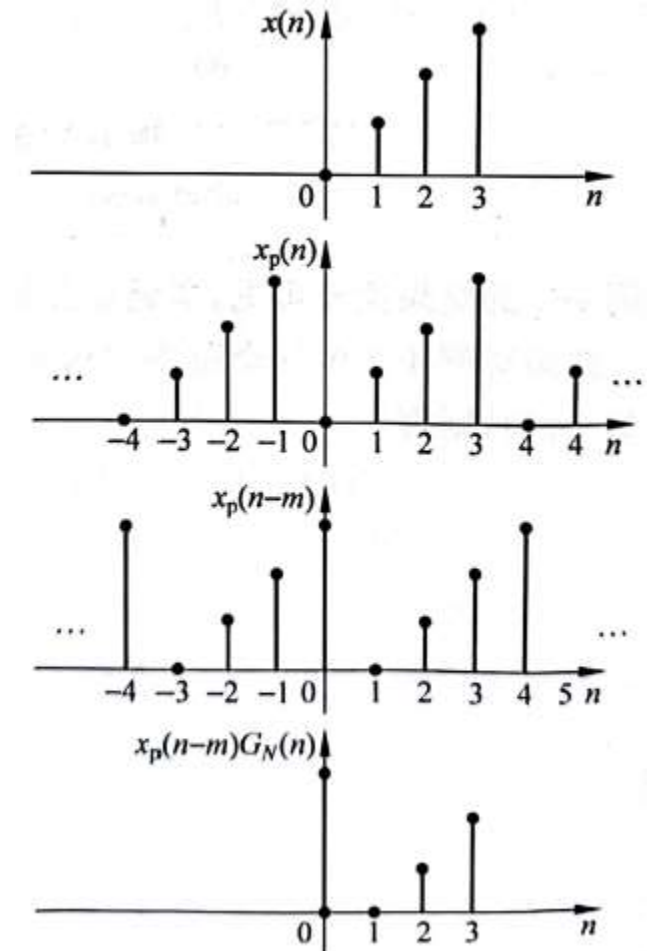
Define a rectangular window

$$G_N(n) = u(n) - u(n - N)$$

Then

$$x(n) = x_p(n)G_N(n)$$

$x_p(n \pm m) \cdot G_N(n)$ represents the sequence obtained by taking the principal interval of a periodic signal after a circular shift of $\pm m$, with a length of N



(2) Circular (Periodic) Shift

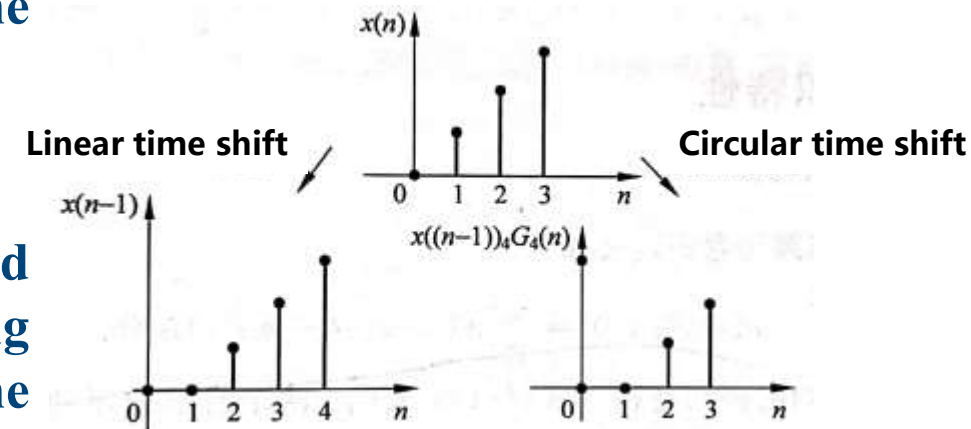
Linear time shift vs. **circular** time shift

Circular time shift:

The interval remains unchanged, the positions shift, and the magnitudes before and after shifting remain the same

Sequential circular shift by m samples:

The N samples of $x(n)$ are arranged around a circle divided into N equal parts (with the start and end connected). Rotating the sequence by m positions around the circle produces the circular shift



$$\{x(n)\} = \{\mathbf{1}, 0, 2, 3\}$$

$$\{x(n-1)\} = \{0, \mathbf{1}, 0, 2, 3\}$$

$$\{x_c(n)\} = \{x_c((n-1))_4 G_4(n)\} = \{3, \mathbf{1}, 0, 2\}$$

(2) Circular (Periodic) Shift

If

$$\text{DFT}\{x(n)\} = X(k),$$

then shifting $x(n)$ circularly by m samples gives:

$$\text{DFT}\{x((n - m))_N\} = W_N^{+mk} X(k),$$

where

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk}, W_N = e^{-j\frac{2\pi}{N}}$$

Usage?

(2) Circular (Periodic) Shift

If

$$\text{DFT}\{x(n)\} = X(k),$$

then a circular shift of the spectrum by m :

$$X_c(k) = X((k - m))_N$$

corresponds in the time domain to:

$$x_c(n) = W_N^{-mn} x(n),$$

where

$$W_N = e^{-j\frac{2\pi}{N}}$$

Usage?

(3) Circular Convolution

Time domain circular convolution of two sequences

$$x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m)h((n-m))_N G_N(n)$$

$\circledast$ Circular Convolutions Representing Two Sequences

Periodically extending and circularly shifting one sequence $h(m)$ with period N , multiplying it with the other sequence $x(m)$, summing the products

The length of sequences $x(m)$ and $h(m)$ should both be N , and the entire convolution process is carried out in the $0 \leq m \leq (N-1)$ interval, with the length $L = N$ after convolution

$((n-m))_N$ denotes $(n-m)$ modulo N , representing the circular shift of the sequence $h(m)$

Example 1

Two finite length sequences are known

$$(N=4)$$

$$x(n) = \delta(n) + \delta(n - 1) + \delta(n - 2)$$

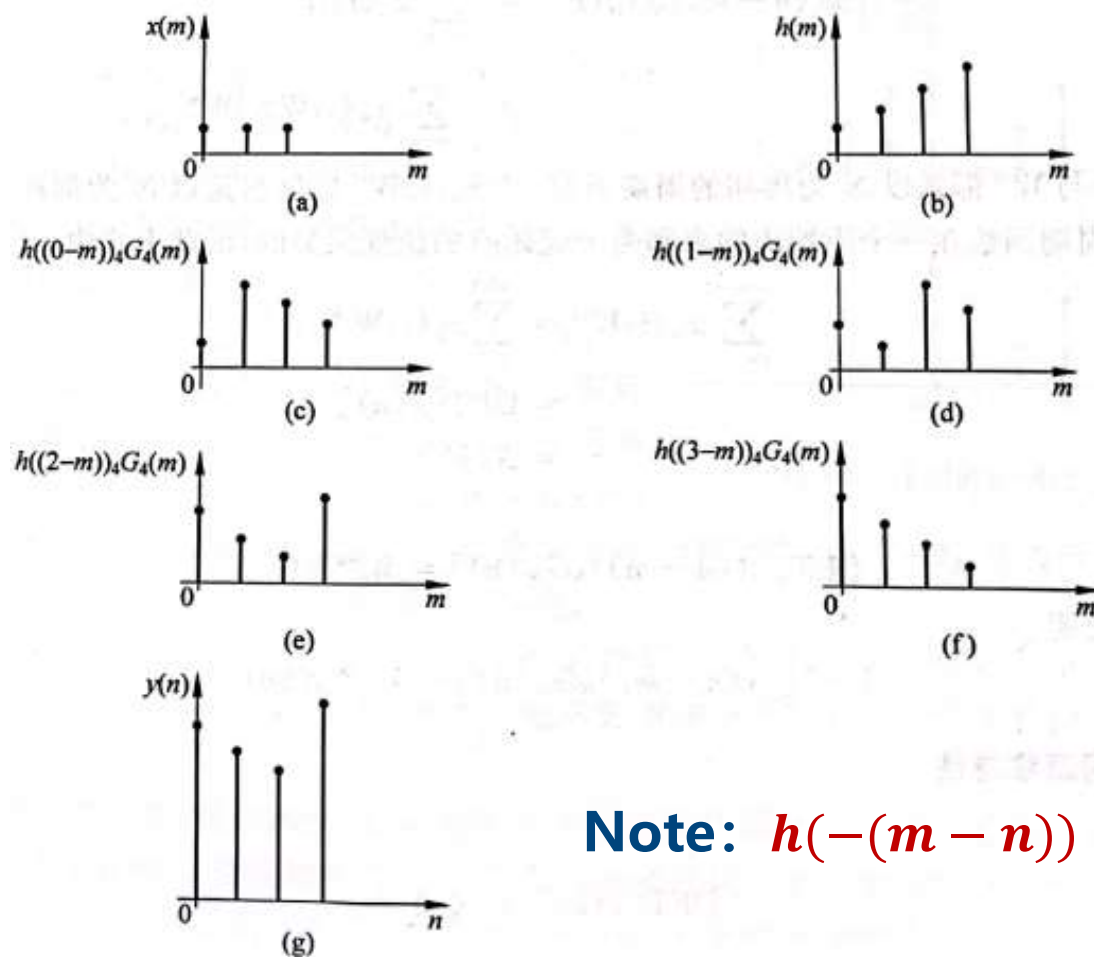
$$h(n) = (n + 1)G_4(n)$$

Graphical method for solving circular convolution

$$y(n) = x(n) \circledast h(n)$$

(3) Circular Convolution

Replace the independent variables n of $x(n)$ and $h(n)$ with m



$$x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m)h((n-m))_N G_N(n)$$

$$y(n) = \sum_{m=0}^3 x(m)h((n-m))_4 G_4(n)$$

$$y(0) = 1 \times 1 + 1 \times 4 + 1 \times 3 + 0 \times 2 = 8$$

$$y(1) = 1 \times 2 + 1 \times 1 + 1 \times 4 + 0 \times 3 = 7$$

$$y(2) = 1 \times 3 + 1 \times 2 + 1 \times 1 + 0 \times 4 = 6$$

$$y(3) = 1 \times 4 + 1 \times 3 + 1 \times 2 + 0 \times 1 = 9$$

obtain,

$$y(n) = 8\delta(n) + 7\delta(n-1) + 6\delta(n-2) + 9\delta(n-3)$$

Note: $h(-(m-n))$

(3) Circular Convolution

Linear convolution vs Circular convolution

$$y(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$$

$$y(n) = x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m)h((n-m))_N G_N(n)$$

(3) Circular Convolution

1. Linear Convolution:

- Sequences $x(n)$ [length N] and $h(n)$ [length M] can have different lengths
- Resulting sequence length: $L = N + M - 1$
- Represents normal time-domain filtering operations

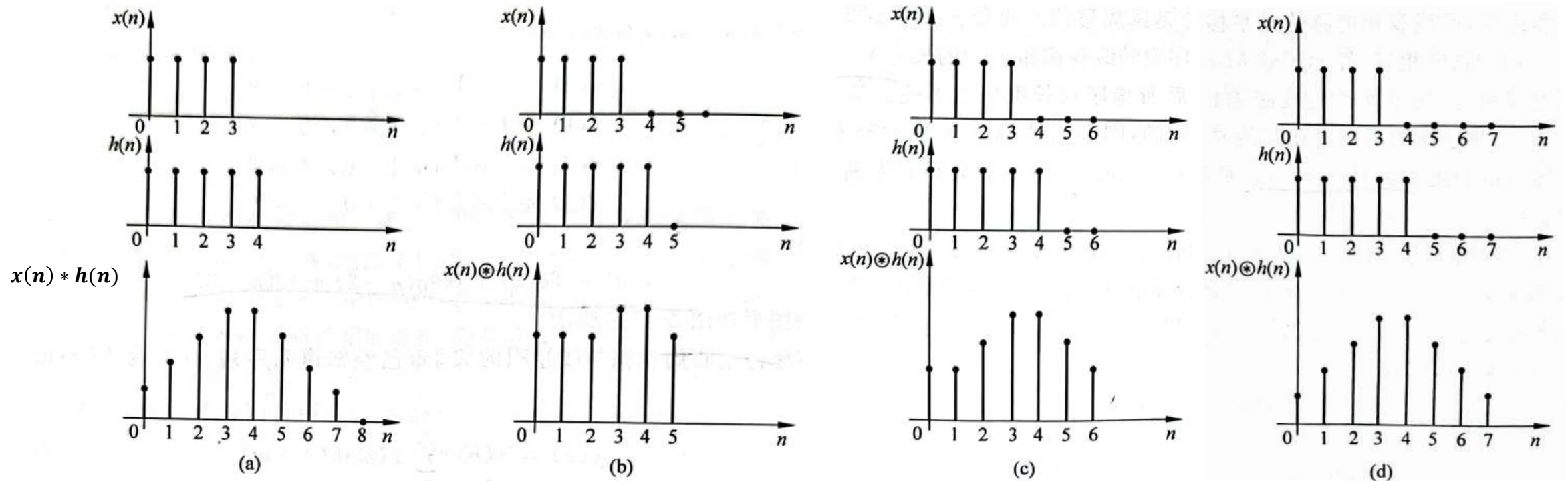
2. Circular Convolution

- Requires both sequences to have equal lengths
- Output sequence maintains the same length as inputs
- Fundamental to DFT operations

To make circular convolution produce **identical results** to linear convolution:

1. Both sequences must be zero-padded to length $L \geq N + M - 1$
2. The padded length L must be identical for both sequences

(3) Circular Convolution



(a) Linear convolution; (b) Circular convolution with $L=6$; (c) Circular Convolution with $L=7$; (d) Circular Convolution with $L=8$

$$L = N + M - 1 = 8$$

$$x(n) * h(n) = x(n) \circledast h(n)$$

(3) Circular Convolution

If

$$\text{DFT}\{x(n)\} = X(k), \text{DFT}\{h(n)\} = H(k),$$

then the circular convolution of $x(n)$ and $h(n)$:

$$y(n) = x(n) \circledast h(n),$$

has the DFT

$$Y(k) = \text{DFT}\{y(n)\} = X(k) H(k)$$

Circular convolution in time corresponds to multiplication in the frequency domain

(3) Circular Convolution

If

$$\text{DFT}\{x(n)\} = X(k), \text{DFT}\{h(n)\} = H(k),$$

and

$$y(n) = x(n) \cdot h(n), Y(k) = \text{DFT}\{y(n)\},$$

then

$$Y(k) = \frac{1}{N} [X(k) \circledast H(k)],$$

where $\circledast$ denotes circular convolution in the frequency domain

Multiplication in the time domain corresponds to circular convolution in the frequency domain (with scaling $1/N$)



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for listening !

